METRIC
MIL-STD-2189(SH)
SECTION 302-1
21 March 1988

MILITARY STANDARD

DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS

SECTION 302-1

AC MOTOR AND CONTROLLER

APPLICATION REQUIREMENTS

AMSC N/A FSC 1990

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Design Methods for Naval Shipboard Systems, Ac Motor and Controller Application Requirements

1. This Military Standard is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. GENERAL AND SCOPE

1.1 General. The procedures established by MIL-STD-2189 are applicable. This section and the basic standard are to be considered as an integral single document.

1.2 Scope. This standard provides methods applicable to typical alternating current (ac) motor-driven auxiliaries proposed for installation aboard ship.

2. REFERNCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standard. Unless otherwise specified, the following specifications and standard of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

A-A-50964 - Grinding Machine, Tool Attachment, Precision.
OO-D-1390 - Dishwashing Machines, Single Tank (Rack, Manual and Conveyor) and Double Tank (Rack, Conveyor), Commercial.
OO-L-125 - Latches, Engine and Toolroom, 10 Inch.
OO-M-38 - Mixing Machines, Food, Electric (Vertical, Commercial Type, Bakery and Kitchen).

MILITARY

MIL-F-68 - Fan, Circulating, Bracket, Nonoscillating, Naval Shipboard.
MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-W-3459 - Wipers, Window, Electric, Pendulum Type (Marine Service, Heavy Duty).
MIL-D-3886 - Dough Divider, Roll, Hand Operated, 36-Part and Dough Divider and Rounding Machine, Roll, 36-Part.
MIL-W-15802 - Winch, Gypsy, Power Operated, Electric.
MIL-F-16081 - Fans, Ventilating, Propeller.
MIL-R-16743 - Refrigerating Plants and Systems, Mechanical; and Refrigeration System Components - Dichlorodifluoromethane Type-12-Naval Shipboard.
MIL-D-17013 - Dough Mixing Machine, Baker's.
MIL-M-17059 - Motors, 60-Cycle, Alternating-Current Fractional H.P. (Shipboard Use).
MIL-M-17060 - Motors, 60-Hertz, Alternating Current Integral-Horsepower (Shipboard Use).

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MILITARY (Continued)

MIL-W-17265 - Winch, Drum, Power Operated (Electric-Hydraulic, Variable Speed); General Specification for.
MIL-M-17312 - Meat Tenderizer, Electric.
MIL-V-17501 - Valves, Globe, Hydraulically Operated, Piston and Diaphragm Type, Magazine and Hanger Deck Sprinkling.
MIL-P-17639 - Pumps, Centrifugal, Miscellaneous Service Naval Shipboard Use.
MIL-E-17807 - Elevator, Weapon and Cargo, Electromechanical (Shipboard).
MIL-P-17840 - Pumps, Centrifugal, Close-Coupled, Navy Standard (For Surface Ship Application).
MIL-P-17869 - Pumps and Motors, Power, Oil Hydraulic (Naval Shipboard Use).
MIL-C-17944 - Capstans, Electric (For Shipboard Use).
MIL-D-18163 - Dry-Cleaning Unit Assembly, Fully Automatic, Synthetic Solvent (Perchlorethylene) Type, (25-35 Pounds Capacity) Shipboard Use.
MIL-C-18419 - Compressors, Reciprocating, Motor Driven, Naval Shipboard High Pressure Air Service.
MIL-P-18472 - Pumps, Centrifugal, Condensate, Feed Booster and Distilling Plant.
MIL-F-18523 - Fans, Centrifugal, Draft, Forced and Induced
MIL-V-18683 - Vacuum Pumps, Power Driven, Centrifugal Pump Priming Service, Naval Shipboard.
MIL-C-19113 - Compressors, Reciprocating, Electric Motor Driven, Naval Shipboard Low Pressure Air Service.
MIL-P-19131 - Pumps, Rotary, Power Driven, Miscellaneous.
MIL-L-19493 - Laundry Pressing and Finishing Equipment Commercial.
MIL-M-19633 - Motor-Generator, 60-Cycle AC to 400-Cycle AC (Voltage and Frequency Regulated) Shipboard Service.
MIL-D-19645 - Display Cases, Mechanically Refrigerated, Self-Contained, Salad and Dessert Dispensing, Naval Shipboard.
MIL-W-19882 - Winches, Drum, Power Operated, Bathythermograph.
MIL-R-21098 - Refrigerators; Frozen Food Cabinets; and Combination Refrigeratior - Frozen Food Cabinets, Mechanically Refrigerated, Commercial, Self-Contained, Naval Shipboard.
MIL-P-22088 - Purifiers, Centrifugal, Jet Fuel, Naval Shipboard.
MIL-C-23122 - Condensers, Air Cooled, Refrigerant-12.
MIL-C-23218 - Conveyor, Vertical, Tray, 100-Pound Package Tray Capacity.
MIL-L-23225 - Lathe, Precision, Instrument.
MIL-L-23257 - Lathes, Engine and Toolroom 1609 and 2012.
MIL-W-23554 - Washer-Extractor, Laundry, Commercial (Shipboard).
MIL-T-23581 - Tumbler, Drying, Laundry, Commercial (Submarine Service).

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MILITARY (Continued)

MIL-S-23639 - System; for Liquid Oxygen or Liquid Nitrogen Storage and Charging (Shipboard Type).
MIL-C-24166 - Cooler, Drinking Water, Mechanically Refrigerated, Self-Contained, Naval Shipboard.
MIL-H-38209 - Hose Reel Assembly, Fuel Servicing, General Requirements for.
MIL-M-40163 - Molding Machines, Meat-Fish.
MIL-F-50084 - Fan, Centrifugal, 12 CFM.
MIL-S-80095 - Saws, Band, Woodworking.
MIL-G-80260 - Grinding Machines, and Grinding and Buffing Machines, Utility, Bench and Pedestal, Electric.
MIL-M-82021 - Monorail Track Beam Systems for Trolley Suspended Hoists.

STANDARD

MILITARY

MIL-STD-2189 - Design Methods for Naval Shipboard Systems.

2.1.2 Other Government publication. The following Government publication forms a part of this standard to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

S9040-AA-IDX-010/SWBS 5D VOL 1 -Expanded Ship Work Breakdown Structure For All Ships and Ship/Combat Systems (ESWBS).

(Copies of specifications, standards and publications required by contractors in connection with specified acqusition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS

3.1 Abbreviations. Abbreviations used in this standard are as follows:

DP	dripproof	OFLRSW	oil failure switch
DPP	dripproof protected	PB	push button
DPST	double-pole single-throw	PRESSW	pressure switch
EP	explosion-proof	REQD	required
EPFC	explosion-proof fan cooled	SBM	submersible
FLTSW	float switch	SPT	spraytight
HPSW	high pressure switch	SPTFC	spraytight fan cooled

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LPSW	low pressure switch	TE	totally enclosed
LIMSW	limit switch	TEFC	totally enclosed fan cooled
LVP	low voltage protection	TSW	temperature switch
LVRLSE	low voltage release	WTRTT	watertight
LVRE	low voltage release effect	WTRTTFC	watertight fan cooled
MAG	magnetic		

4. GENERAL REQUIREMENTS

4.1 Use of table I. The "AC motor design" and "Navy service" columns in table I reflect the designs and classes specified in MIL-M-17059 and MIL-M-17060. The definitions of LVP and LVRLSE are as specified in MIL-C-2212. Table I is provided as a guide and does not modify the requirements of specifications for auxiliaries, or the requirements for the location or application of any specific installation. The time ratings of varying, intermittent, and short-time motors shall be as specified in table I, unless there are sufficient tests or service data available to show that a different time rating more closely meets the requirements of the specific application. For additional and more specific information on equipment requirements and the application of motors and controllers, refer to applicable ESWBS sections, particularly sections 300 and 302.

5. DETAILED REQUIREMENTS (Not applicable)

6. NOTES

6.1 Intended use. This standard is intended for use as a guide in selecting motors, controllers, and master switches.

6.2 Subject term (key word) listing.

Dripproof
Explosion-proof
Float switch
High pressure switch
Low pressure switch

Preparing activity:
Navy - SH
(Project 1990-N054)

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TABLE I. Ac motor and controller application requirements.

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Auxiliary propulsion:											
Auxiliary propulsion unit	Squirrel cage/ constant	B	Continuous	SBM	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Retracting system	Squirrel cage/ constant	D	Short time, 1/2 hour	TE	A	Starting, reversing	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Training system	Squirrel cage/ constant	D	Short time, 1/2 hour	TE	A	Starting, reversing	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Bow thruster	Squirrel cage/ constant	B	Contin-	SBM	A	Starting	MAG, LVP	Nonauto-	DP	Local, remote PB	
Capstan, warping	Squirrel cage/ mulitspeed	C	Varying, 1/2 hour	DPP, WTRTT	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	DP or WTRTT	Remote, WTRTT drum type emergency remote PB	MIL-C-17944
Capstan, mooring	Squirrel cage/ multispeed	C	Varying, 1/2 hour	DPP, WTRTT	A	Starting, reversing speed selection	MAG, LVP	Nonautomactic	DP or WTRTT	Remote, WTRTT drum type emergency remote PB	MIL-C-17944

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Cleaners, vacuum	Universal/ contant	-	Continuous	DPP	C	Starting	Manual	-	-	-	
Compressor:											
Air conditioning	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic and automatic	DP	Local TSW PRESSW and PB	MIL-C-23122 MIL-R-16743
Air, deballast	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Automatic	DP	Local and remote PB	
Air, diesel starting	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic and automatic	DP	Local, remote HPSW, PRESSW LPSW, TSW	MIL-C-18419
Air, diver	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	-	
Air, high pressure	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Automatic	DP	Local PB HPSW, LPSW TSW	MIL-C-18419

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Compresser (continued):											
Air, incinerator	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Air, laundry	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Air, low pressure	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Automatic, nonautomatic	DP	Local remote PB, HPSW, LPSW TSW, OFLRSW	MIL-C-19113
Air, sewage holding tank	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB, LIMSW	
Air, ship service	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Automatic	DP	Remote PB HPSW, LPSW	
O_2N_2 plant	Squirrel cage/ constant	B	Continuous	DPP, STP	A	Starting	MAG, LVP	Semiautomatic	DP, WTRTT	Local PB, PRESSW	MIL-S-23639

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Compressor (continued):											
Prairie masker	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Refrigeration dispenser, canned juice and ice cream	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW	
Refrigeration drinking water coolers	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW	MIL-C-24166
Refrigeration frozen food cabinet	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW	MIL-R-21098
Refrigeration ice cream batch freezer	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVRLSE	Automatic	DP	Local PB, TSW, HPSW, LPSW	
Refrigeration ice making machine automatic	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW HPSW	

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Compressor (continued):											
Refrigeration Refrigerator	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW	MIL-R-21098
Refrigeration refrigerator-freezer combination	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW	MIL-R-21098
Refrigeration salad-dessert dispenser bar	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local LPSW, TSW	MIL-D-19645
Refrigeration ship stores refrigerating unit/reciprocating unloaded start	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Automatic	DP	Local or remote TSW PRESSW and PB	MIL-R-16743
Refrigeration ship stores refrigerating unit/reciprocating loaded start	Squirrel cage/ single or multispeed	C	Continuous	DPP	A	Starting speed selection	MAG, LVP	Nonautomatic, automatic	DP	Local TSW PRESSW and PB	MIL-R-16743

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Compresser (continued):											
Refrigeration soft ice cream freezer	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local TSW	
Refrigeration under-counter refrigerator	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	Local HPSW and TSW	
Conveyers:											
Pallet and package	Squirrel cage/ constant	B,C	Continuous	As Reqd	C	Starting, reversing	MAG, LVP	Nonautomatic	As Required	Local, remote PB	
Stores, vertical	Squirrel cage/ constant	B	Continuous	DPP	C	Starting, reversing	MAG, LVP	Nonautomatic	DP	ON-OFF switch	MIL-C-23218
Crane:											
Boat or aircraft (electro-hydraulic)	Squirrel cage/ constant	B	Varying, 1/2 hour	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	

See footnote at end of table.

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Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Crane (continued):											
Boat or aircraft (electric)	Squirrel cage	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Jib crane	Squirrel cage/ multispeed	D	SP	WTRTT	A	Starting, reversing	MAG, LVP	Nonautomatic	DP	Remote PB (SBM)	
Curtain, roller	Squirrel cage/ varying speed	C	Short time, 1/2 hour	SPT, SPTFC	A	Starting reversing speed selection	MAG, LVP	Nonautomatic		Local, remote PB	Limit switches, as required
Davit, rotating gear	Squirrel cage/ varying speed	D	Short time, 1/2 hour	WTRTT	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	DP	Local, remote PB	Limit switches and interlocks, as required
Door, bow ramp (electro-hydraulic	Squirrel cage/ constant	B	Varying, 1/2 hour	DPP	A	Starting	MAG, LVP	Nonautomatic	WTRTT	Local, remote PB	Limit switches and interlocks, as required
Door, motor operated	Squirrel cage/ multispeed	D	Short time 1/2 hour	As required	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	As required	Remote PB	Limit switches and interlocks, as required

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Elevator:											
Aircraft (electro-hydraulic)	Squirrel cage/ multispeed	B,C	Varying, 1/2 hour	DPP	A	Starting, speed selection	MAG, LVP	Nonauto-matic	As re-quired	Remote PB	MIL-P-17869
Aircraft (electric)	Squirrel cage/ varying, multispeed	D	As re-quired	SPT, SPTFC	A	Starting, reversing speed selection	MAG, LVP	Nonauto-matic	As re-quired	Remote PB	MIL-E-17807
Bomb (elec-trohydrau-lic)	Squirrel cage/ constant	B	Varying, 1 hour	As re-quired	A	Starting	MAG, LVP	Nonauto-matic	DP	Remote PB	MIL-P-17869
Bomb, or tor-pedo (elec-tric)	Squirrel cage/ varying, multispeed	D	As re-quired	As re-quired	A	Starting, reversing speed selection	MAG, LVP	Nonauto-matic	DP	Remote drum or PB	MIL-E-17807
Cargo, main drive	Squirrel cage/ multispeed	D	Inter-mittent	DPP	A	Starting, reversing speed selection	MAG, LVP	Nonauto-matic	As re-quired	Remote PB	
Pallet	Squirrel cage/ multispeed		Varying, 1 hour	DPP	A	Starting, speed selection	MAG, LVP	Nonautop matic	DP	Local, remote PB	

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Elevator: (continued):											
Screw type	Squirrel cage/ varying, multispeed	D	As required	As required	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	DP	Remote PB	
Strike down	Squirrel cage/ varying multispeed	D	As required	As required	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	DP	Remote PB	
Small arms	Squirrel cage/ varying, multispeed	D	As required	As required	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	DP	Remote PB	
Fans:											
Air, primary	Squirrel cage/ constant	B	Continuous	WTRTT	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Bracket	Shaded coil/ varying, multispeed	-	Continuous	DP, DPP	C	Starting	Manual	-	-	-	MIL-F-68

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Fans (continued):											
Centrifugal	Squirrel cage/ multispeed constant	B	Contin-uous	TE	A	Starting or stop-ping speed selection	MAG, LVP	Nonauto-matic	As re-quired	Local PB	MIL-F-50084
Circulating, ceiling (ref-rigerator spaces)	Shaded coil/ varying	-	Contin-uous	DP	C	Starting	Manual	-	-	-	
Forced draft main or port use (centri-fugal or vaneaxial)	Squirrel cage/ multispeed	B	Contin-uous	DPP	A	Starting, speed selection	MAG, LVP	Nonauto-matic	As re-quired	Local, remote PB	
Gland exhaust	Squirrel cage/ constant	B	Contin-uous	SPTFC	A	Starting	MAG, LVRLSE	Nonauto-matic	As re-quired	Local PB	
Main engine cooling	Squirrel cage/ multispeed	B	Contin-uous	SPT	A	Starting, speed selection	MAG, LVRLSE	Automatic and non-automatic	DP	Local, remote PB	

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Fans (continued):											
Portable	Split phase/ constant	B	Continuous	EXP	A	Starting	Manual	-	As required	Local PB	
Ship service/ emergency gas turbine generator enclosure cooling	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVRLSE	Automatic	DP	Local, remote PB	MIL-F-18523
Tubeaxial or propeller type	Multispeed or constant	B	Continuous	EP SPT	A	Starting	MAG, LVRLSE	Nonautomatic	As required	Local PB	MIL-F-16081
Unit cooler	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	Manual, LVRE	-	DP	-	
Vaneaxial type	Squirrel cage/ two-speed or constant	B	Continuous	EP, SPT	A	Starting speed selection	MAG, LVP	Nonautomatic	As required	Local PB	
Food service equipment:											

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Food service equipment (continued):											
Dumbwaiter	Squirrel cage/ varying	D	Intermittent, 1/2 hour	TE	C	Starting, reversing	MAG, LVP	Nonautomatic	DP	Remote PB	
Garbage grinder	Squirrel cage/ constant	B	Intermittent, 1 hour	SPT, SPTFC	C	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Knife sharpener	Squirrel cage/ constant	B	Short time, 1/4 hour	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	PB	
Machine bread-slicing	Capacity start/ constant	-	Continuous	DPP or TE	C	Starting	Manual	-	DP	-	
Machine, chopping (grinding) meat	Squirrel cage/ constant	B	Continuous	DPP or TE	C	Starting	Manual	-	DP	-	(0.4 kW (larger than 1/2 HPSW))
Machine, chopping (grinding) meat	Capacity start/ constant	-	Continuous	DPP or TE	C	Starting	Manual	-	DP	-	(0.4 kW (1/2 HPSW and less))

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Foodservice equipment (continued):											
Machine, dishwashing	Squirrel cage/ constant	B	Continuous	TE	C	Starting	MAG, LVP	Semi-automatic	DP	Local PB	OO-D-1390
Machine, dough divider	As required	B	Intermittent, 1/2 hour	DPP	C	Starting	Manual	-	DP	-	MIL-D-3886
Machine, dough mixing	Squirrel cage/ constant	B	Continuous	SPT, SPTFC	C	Starting	MAG, LVP	Nonautomatic	DP	Remote	MIL-D-17013
Machine, food mixing	Squirrel cage/ constant	-	Varying, 1 hour	DPP, TE or TEFC	C	Starting	Manual	-	DP	DPST switch	OO-M-38 (less than (0.4 kW (1/2 HPSW))
Machine, food mixing	Squirrel cage/ constant	B	Varying, 1 hour	DPP, TE or TEFC	C	Starting	MAG, LVP	Nonautomatic or semi-automatic	DP	Local PB	OO-M-38 (0.4 kW (1/2 HPSW and larger))
Machine, food shaping	Squirrel cage/ constant	B	Continuous	TE	C	Starting	Manual	-	DP	-	MIL-M-40163

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Food service equipment (continued):											
Machine, meat, bone and fish cutting	Squirrel cage/ constant	B	Continuous	TE, TEFC	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Machine, meat tenderizing	Squirrel cage/ constant	B	Continuous	TE	C	Starting	Manual	-	DP	-	MIL-M-17312
Machine, slicing, meat	Capacity start constant	-	Continuous	TE, DPP	C	Starting	Manual	-	DP	-	
Machine, vegetable, peeling	Squirrel cage/ constant	B	Varying, 1 hour	SPT, SPTFC	C	Starting	Manual	-	WTRTT	-	
Machine, vegetable slicer	Capacity start/ constant	-	Varying, 1 hour	TE, DPP	C	Starting	Manual	-	DP	-	
Toaster conveyor	Squirrel cage/ constant	B	Varying	TE, DPP	C	Starting	Manual	-	DP	-	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Foodservice equipment (continued):											
Trash compactor	Squirrel cage/ constant	B	Intermittent 1 hour	SPT, SPTFC	C	Starting	Manual	-	DP	-	
Hoists:											
Accomodation lader	Squirrel cage/ constant	D	Continuous	WTRTT	A	Starting reversing	MAG, LVP	Nonautomatic	DP	Remote PB (SBM)	
Ammunition (hydraulic dredger)	Capacity start/ constant	B	Varying, 1/2 hour	SPT, SPTFC	A	Starting	MAG, LVP	Nonautomatic	WTRTT	Local PB	
Cargo (hanger)	Squirrel cage/ multispeed	D	Intermittent	DPP	A	Starting, reversing speed, selection	MAG, LVP	Nonautomatic	DP	Remote PB (SBM)	
Hose reel	Squirrel cage/ constant	B	Continuous	SPTFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	MIL-H-38209

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Hoists (continued):											
Landing craft line delivery system	Squirrel cage/ constant	D	Intermittent	WTRTTFC	A	Starting, reversing	MAG, LVRLSE	Nonautomatic	WTRTT	Remote PB (SBM)	
Monorail	Squirrel cage/ varible	C	Intermittent	DPP	A	Starting, reversing speed selection	MAG, LVP	Nonautomatic	DP	Remote PB (SBM)	MIL-M-82021
Wire rope and chain (hatch, ammunition, and stores)	Capacity start/ constant	C	Short time, 1/2 hour	SPT, WTRTT	A, C	Starting, reversing	MAG, LVP	Nonautomatic	WTRTT	Local PB	
Laundry equipment:											
Dry-cleaning outfit	Squirrel cage/ constant	As required	Varying, 2 hours	TE, TEFC	C	Starting or starting and reversing	MAG, LVP	Automatic	DP	Local PB	MIL-D-18163
Ironer, flatwork	Squirrel cage/ constant	B	Continuous	TE	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	MIL-L-19493

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Laundry equipment (continued):											
Shaper	Squirrel cage/ constant		Contin-uous	TEFC	C	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Tumbler, dryer, drum	Squirrel cage/ constant	B	Contin-uous	TE	C	Starting	Manual	-	-	-	MIL-L-19493
Tumbler, drying (submarines)	Squirrel cage/ constant	-	Contin-uous	TE, SPT	C	Starting reversing	MAG, LVP	Automatic	DP	Local PB	MIL-T-23581
Washer, extractor	Squirrel cage/ constant	FF	Inter-mittent	TE, TEFC	C	Starting, reversing	MAG, LVP	Automatic	DP	Local PB	MIL-L-19493
Washer, small type	Squirrel cage/ constant	-	Contin-uous	TE, TEFC	C	Starting, reversing	MAG, LVP	Automatic	DP	Local PB	
Washer, heavy duty (submarines)	Squirrel cage/ constant	B	Contin-uous	TE, SPT	C	Starting, reversing	MAG, LVP	Automatic	DP	Local PB	MIL-W-23554

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Motor generator:											
Arc welding	Squirrel cage/ constant	B	Continuous, 60 percent duty cycle	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Interior communication	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Ship service and special frequency supply	Squirrel cage/ constant	B	Continuous	DPP TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	MIL-M-19633
Printing and lithographic equipment:											
Offset press	Squirrel cage/ varying	B	Continuous	DPP	C	Starting, reversing, inching	MAG, LVP	Nonautomatic	DP	Remote PB	43 X 56 cm (17 X 22 inch) and 56 X 86 cm (22 X 34 inch)
Offset press	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	36 X 51 cm (14 X 20 inch)

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Printing and lithographic equipment (continued):											
Paper cutting machine	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	91-cm (36-inch)
Plate-making machine	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	-	
Temperature controlled sink	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVRLSE	Automatic	DP	Local TSW	
Vacuum pump (for camera)	Squirel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	-	
Wire stitcher	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	Manual	-	DP	-	
Pumps:											
A/C, chilled water	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (Continued):											
A/C, plant purge	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Aqueous film forming foam injection	Sqirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVR	Nonautomatic	DP	Local and remote PB	
Auxiliary boiler feed	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	As required	DP	As required	
Auxiliary circulate	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Remote PB	
Auxiliary condensate	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Remote PB	
Auxiliary machinery cooling water service	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	MIL-P-17840

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Auxiliary vacuum	Squirrel cage/ constant	B	Continuous	WTRTTFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Remote PB	
Asroc fluid circulating	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	Manual, LVRE	-	DP	Local PB	
Bilge	Squirrel cage/ constant	B	Continuous	SBM	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Bromine treatment	Squirrel cage/ varying	B	Continuous	TEFC	C	Starting	MAG, LVP	Nonautomatic	WTRTT	Local PB, PRESSW	
Boiler burner blower and fuel	Squirrel cage/ varying	B	Continuous	DPP, TEFC	A	Starting	MAG, LVRLSE	Automatic	TE	Local PB	
Centrifugal pump priming (non-flammable)	Squirrel cage/ constant	B	Short time	DPP, TEFC	A	Starting	MAG, LVP	Nonautomatic, automatic	DP	Local, remote PB PRESSW	MIL-V-18683

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
CO_2 recharge	Squirrel cage/ constant	B	As required	DPP	C	Starting	Manual	-	-	-	
Continuous wave illuminator (CWI) circulating	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Controllable reversible pitch propeller hydraulic	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP, LSE	Automatic and nonautomatic	DP	Local, remote PB	
Demineral water circulate	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB and PRESSW	
Distilling plant pumps:											
Booster	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Distilling plant pumps (continued):											
Brine over-board	Sqirrel cage/ constant	B	Contin-uous	DPP, TE	A	Starting	MAG, LVP	Nonauto-matic	DP	Local, remote PB	
Circulating	Squirrel cage/ constant	B	Contin-uous	TE, TEFC	A	Starting	MAG, LVP	Nonauto-matic	DP	Local, remote PB	
Condensate	Squirrel cage/ constant	B	Contin-uous	TE, TEFC	A	Starting	MAG, LVP	Nonauto-matic	DP	Local, remote PB	MIL-P-18472
Distiller distillate	Squirrel cage/ constant	B	Contin-uous	TEFC	A	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	MIL-P-17840
Distiller SW feed	Squirrel cage/ constant	B	Contin-uous	TEFC	A	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Distiller heater drain	Squirrel cage/ constant	B	Contin-uous	TEFC	A	Starting	MAG, LVP	Nonauto-matic	-	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Distilling plant pumps (continued):											
Distiller plant chemical feed	Squirrel cage/ constant	B	Continuous	DPP, TEFC	C	Starting	MAG, LVRLSE	Nonautomatic	DP	Local PB	
Evaporator feed	Squirrel cage/ constant	B	Continuous	SPT, SPTFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Tube nest drain	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	
Educter supply	Squirrel cage/ constant	B	Continuous	TE, WTRTT	A	Starting	Manual	-	DP	-	
Electronic cooling and chilled water	Squirrel cage/ constant	B	Continuous	TE	A	Starting	MAG, LVP	Nonautomatic	DP	PB	
Elevator oil supply	Squirrel cage/ constant	B	Short time, 1 hour	DPP	A	Starting	MAG, LVP	Automatic	DP	Remote PB FLTSW	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Pumps (continued):											
Elevator, aircraft, main	Squirrel cage/ constant	B	Contin-uous	DPP	A	Starting	MAG, LVP	Automatic	DP	Local PB	
Emergency, booster and reserve feed transfer	Squirrel cage/ constant	B	Contin-uous	DPP	A	Starting	MAG	Nonauto-matic, automatic	DP	Local, remote PB PRESSW	
Fire	Squirrel cage/ constant	B	Contin-uous	TEFC	A	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Fresh water drain collec-ting tank	Squirrel cage/ constant	B	Contin-uous	TE	A	Starting	MAG, LVP	Nonauto-matic	DP	PB	
Fresh water service	Squirrel cage/ constant	B	Contin-uous	TEFC	A	Starting	MAG, LVP	Automatic	DP	Local, remote PB PRESSW	
Fresh water booster	Squrrel cage/ constant	B	Contin-uous	TEFC	A	Starting	MAG, LVP	Automatic	DP	Local PB, PRESSW	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Fuel-oil service, port and cruising	Squirrel cage/ multispeed	B	Continuous	TE, TEFC	A	Starting, speed selection	MAG, LVP	Nonautomatic	DP	Remote PB	
Fuel-oil tank drain	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Fuel-oil transfer	Squirrel cage/ multispeed	C	Continuous	TE, TEFC	A	Starting, speed selection	MAG, LVP	Nonautomatic	TE	Local PB	
Gasoline defueling	Squirrel cage/ constant	B	Continuous	EP, EPFC	A	Starting	MAG, LVP	Nonautomatic	EP	Local, remote PB (EB)	
Gas turbine wash module	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Local, remote PB	
Hot fresh water recirculating	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Hot water circulating	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Incinerator feedwater cooling	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Incinerator ignition oil	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	Manual	-	TE	-	
Incinerator oil	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	Manual	-	TE	-	
Infrared (IR) suppression	Squirrel cage/ constant	B	Continuous	DPP	A	Starting,	MAG, LVP	Nonautomatic	DP	Local PB	
Injection, 76mm gun	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Incinerator, rotating grate	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	Manual	-	DP	-	
JP-5 auxiliary service	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	TE	Local, remote PB	
JP-5 defueling	Squirrel cage/ constant	B	Continuous	TE TEFC	A	Starting	MAG, LVP	Nonautomatic	TE	Local PB	MIL-P-19131
JP-5 priming	Squirrel cage	B	Short time, 3 minutes	TE, TEFC	A	Starting, dual control	MAG, LVP	Nonautomatic	TE	Local, remote PB	MIL-V-18683
JP-5 service	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	TE	Local PB	MIL-P-17639
JP-5 stripping	Squirrel cage/ constant	B	Short time, 3 minutes	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	TE	Local PB	MIL-P-19131

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Pumps (continued):											
JP-5 transfer	Squirrel cage/ constant	B	Contin-uous	TE, TEFC	A	Starting	MAG, LVP	Nonauto-matic	TE	Local PB	MIL-P-19131
Low pressure steam drain transfer	Squirrel cage/ constant	B	Contin-uous	TE	A	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	MIL-P-18472
Lube-oil	Squirrel cage/ multispeed	B	Contin-uous	TEFC	A	Starting speed selection	MAG, LVRLSE	Nonauto-matic	DP	Remote PB	
Lube-oil F.D. fan	Squirrel cage/ constant	D	Contin-uous	TE	A	Starting	MAG, LVP	Nonauto-matic	DP	Remote PB	
Lube-oil pump turbine-driven fire pump	Squirrel cage/ constant	B	Contin-uous	TE	A	Starting	MAG, LVRLSE	Nonauto-matic	DP	Remote PB	
Lube-oil pump main, feed pump	Squirrel cage/ constant	B	Varying	TE	A	Starting	MAG, LVRLSE	Nonauto-matic	DP	Remote PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Lube-oil pump main, service	Squirrel cage/ multispeed	B	Continuous	TEFC	A	Starting, speed selection	MAG, LVRLSE	Nonautomatic	DP	Remote PB	
Lube-oil, turbine generator set (STBY)	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVRLSE	Automatic	DP	Remote PB and PRESSW	
Macerator sewage	Squirrel cage/ constant	B	Continuous	TE, TEFC	C	Starting, reversing	MAG, LVP	Automatic and nonautomatic	DP	Local PB, LIMSW	
Machinery waste water drain	Squirrel cage/ constant	B	Continuous	TEFC, WTRTT	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB, LIMSW	
Main condenser condensate	Squirrel cage/ constant	B	Continuous	WTFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Remote PB	
Main condenser circulating	Squirrel cage/ constant	B	Continuous	DPP	A	Starting, speed	MAG, LVP	Nonautomatic	DP	Local, remote PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Main feed booster	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Local PB	
Mogas service	Squirrel cage/ constant	B	Continuous	EPFC	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Mogas stripping	Squirrel cage/ constant	B	Continuous	EPFC	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Mogas salt water service	Squirrel cage/ constant	B	Continuous	EPFC	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	
Portable submersible	Squirrel cage/ constant	B	Continuous	SBM (15 ft) 5 m.	A	Starting	Manual LVRE	-	WTRTT	-	
Potable water	Squirrel cage/ constant	B	Continuous	TEFC, SPT, WTRTT, SPTFC, WTRTTFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB, PRESSW	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Portable water priming	Squirrel cage/ varying	-	Continuous	TE	A	Starting	MAG, LVP	Automatic	DP	Local PB, vacuum SW	
Refrigerator condenser circulating	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Reserve feed transfer	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Sea water booster (emergency generator)	Squirrel cage/ constant	B	Continuous	SBM	A	No controller required	-	-	-	-	MIL-P-17639
Sea water circulating, main propulsion	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Local, remote PB	Interlock switch with suction and discharge valves
Sea water, service	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	

See footnote at end of table.

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Sea water booster (gasoline system)	Squirrel cage/ constant	B	Continuous	EPFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	MIL-P-17639
Sonar cooling	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local, remote PB	
SSTG condenser circulating	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting, speed selection	MAG, LVRLSE	Nonautomatic	DP	Local PB	MIL-P-17840
SSTG condenser condensate	Squirrel cage/ constant	B	Continuous	TE	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Local PB	MIL-P-18472
Sewage	Squirrel cage/ constant	B	Continuous	TEFC	C	Starting	MAG, LVP	Automatic and nonautomatic	DP	Remote PB LIMSW	
Sewage comminutor	Squirrel cage/ constant	B	Continuous	TEFC	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Pumps (continued):											
Sewage ejector	Squirrel cage/ constant	B	Continuous	TE	A	Starting	MAG, LVP	Semiautomatic	DP	Local, remote PB LIMSW	
Vacuum, main condenser	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVRLSE	Nonautomatic	DP	Remote PB	
Waste heat, hot water circulating	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVRLSE	Automatic manual	DP	Local PB, PRESSW	
Waste water bilge sump tank	Squirrel cage/ constant	B	Continuous	TE	A	Starting	MAG, LVP	Automatic	DP	FLTSW	
Water barrier	Squirrel cage/ constant	D	Intermittent	WTRTT	A	Starting, reversing	MAG, LVP	Nonautomatic	SBM	Local, remote PB (SBM)	
Purifiers:											
Diesel fuel oil	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Purifiers (continued):											
JP-5	Squirrel cage/ constant	B	Continuous	TEFC	A	Starting	MAG, LVP	Semiautomatic	DP	Local PB	MIL-P-22088
Main lubricating oil	Squirrel cage/ constant	B	Continuous	TE, TEFC	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Shaft turning gear	Squirrel cage/ varying high slip	D	Continuous	DPP	A	Starting, reversing	MAG, LPV	Nonautomatic	DP	Local PB	
Shop equiment:											
Arbor press	Squirrel cage/ constant		Short time	DPP	C	Starting	Manual	-	DP	Switch	
Armature coil	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Bench grinder	Squirrel cage/ constant	B	Varying 1 hour	SPTFC WTRTT, WTRTTFC	C	Starting	MAG, LVP	Nonautomatic	WTRTT	Local PB	MIL-G-80260

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Shop equiment (continued):											
Bench lathe	Squirrel cage/ constant	B	Contin-uous	DPP, TE	C	Starting, reversing	MAG, LVP	Nonauto-matic	DP	Local PB	00-L-125
Bending and crimping machine	Squirrel cage/ varying	D	Varying, 1 hour	DPP	C	Starting	MAG, LVP	Nonauto-matic	WTRTT	Local PB	
Boring mill	Squirrel cage/ constant	B	Contin-uous	DPP, TE	C	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Buffer	Squirrel cage/ constant	B	Varying, 1/2 hour	SPT, SPTFC, WTRTT, WTRTTFC	C	Starting	MAG, LVP	Nonauto-matic	WTRTT	Local PB	MIL-G-80260
Buffer (oil burner cleaner	Squirrel cage/ constant	B	Varying 1/2 hour	SPT, SPTFC, WTRTT, WTRTTFC	C	Starting	MAG, LVP	Nonauto-matic	WTRTT	Local PB	
Coil winding machine	Squirrel cage/ constant	B	Contin-uous	DPP	C	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Shop equipment (continued):											
Combination machine	Squirrel cage/ constant	B	Intermittent 1 hour	TE, TEFC	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Crank planer and shaper	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Double emery grinder	Squirrel cage/ constant	B	Varying, 1 hour	TE, SPTFC, WTRTT, WTRTTFC	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Drill press	Squirrel cage/ constant	B	Varying, 1 hour	TE	C	Starting	MAG, LVP	Nonautomatic	DP	Local PB	
Engine lathe	Squirrel cage/ constant	B	Continuous	DPP	C	Starting, reversing	MAG, LVP	Nonautomatic	DP	Local PB	MIL-L-23257
Gap lathe	Squirrel cage/ constant	B	Continuous	DPP	C	Starting, reversing	MAG, LVP	Nonautomatic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Shop equipment (continued):											
Grinder, aerial	Universal varying	B	Varying, 1/2 hour	TE	C	Starting	Manual	-	WTRTT	Switch	
Grinder, pedestal	Squirrel cage/ constant	B	Contin-uous	TE	C	Starting	Manual	-	WTRTT	Switch	
Hammer	Squirrel cage/ constant	D	Contin-uous	DPP	C	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Internal grinder	Squirrel cage/ constant	B	Contin-uous	SPT SPTFC	C	Starting	MAG, LVP	Nonauto-matic	WTRTT		
Jeweler's lathe	Universal varying	-	Varying, 1 hour	DPP	C	Starting	MAG, LVP	Nonauto-matic	DP	Foot switch	MIL-L-23225
Key cutting machine	Squirrel cage/ constant	B	Inter-mittent, 1/2 hour	DPP	C	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Machine forming	Squirrel cage/ constant	B	Varying, 1 hour	DPP	C	Starting	MAG, LVP	Nonauto-matic	TE	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Shop equipment (continued):											
Metal cutter, sheet	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP		
Nibbling machine	Squirrel cage/ constant	B	Varying, 1 hour	DPP	C	Starting	MAG, LVP	Nonautomatic	WTRTT		
Planer and joiner	Squirrel cage/ constant	B	Varying, 1 hour	TE	C	Starting	MAG, LVP	Nonautomatic	WTRTT		
Portable drill	As required/ varying	-	Intermittent, 15 minutes	DPP	C	Starting	Manual	-	DP	-	
Power hacksaw	Squirrel cage/ constant	B	Intermittent, 1 hour	DPP	C	Starting	MAG, LVP	Nonautomatic	WTRTT	Local PB	
Precision lathe	Squirrel cage/ varying	B	Continuous	SPT, SPTFC, WTRTT, WTRTTFC	C	Starting, reversing	MAG, LVP	Nonautomatic	DP	Foot switch	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Shop equipment (continued):											
Radial drill	Squirrel cage/ constant	B	Varying, 1/2 hour	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	-	
Routing machine	Squirrel cage/ constant	B	Varying, 1 hour	DPP	C	Starting	Manual	-	DP	-	
Sander belt and finish	Squirrel cage/ constant			TE	C	Starting	Manual	-	WTRTT	-	
Saw, band	Squirrel cage/ constant	B	Varying, 1 hour	TEFC	C	Starting	MAG, LVP	Nonautomatic	WTRTT	-	MIL-S-80095
Saw, circular	Squirrel cage/ constant	B	Varying, 1 hour	TE	C	Starting	MAG, LVP	Nonautomatic	WTRTT	-	30 cm (12-inch)
Sawdust and shavings collecting system blower	Squirrel cage/ constant	B	Continuous	TE	C	Starting	MAG, LVP	Nonautomatic	DP	-	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Shop equipment (continued):											
Sensitive drill (floor type)	Squirrel cage/ multispeed	B	Varying, 1 hour	DPP	C	Starting, speed selection	MAG, LVP	Nonautomatic	DP	-	
Sewing machine	Squirrel cage/ constant	-	Continuous	DPP	C	Starting	Manual	-	DP	-	
Shear, portable	Universal/ varying	-	Continuous	DPP	C	Starting	Manual	-	DP	-	
Shoe finishing machine	Squirrel cage/ constant	B	Continuous	TE	C	Starting	Manual	-	DP	-	
Shoe repair machine	Capacitor start/ multispeed	B	Continuous	DPP	C	Starting, speed selection	Manual	-	DP	-	
Tool post grinder (portable)	Squirrel cage/ constant	B	Continuous	TEFC	C	Starting	Manual	-	DP	-	A-A-50964
Universal cutter and tool grinder	Squirrel cage/ constant	B	Continuous	SPT, SPTFC	C	Starting	MAG, LVP	Nonautomatic	DP	-	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Shop equipment (continued):											
Universal milling machine	Squirrel cage/ constant	B	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	-	
Valve refacer	Universal/ varying	-	Continuous	DPP	C	Starting	MAG, LVP	Nonautomatic	DP	-	
Woodworking lathe	Squirrel cage/ multispeed	B	Continuous	DPP	C	Starting, speed selection	MAG, LVP	Nonautomatic	WTRTT	Local PB	
Woodworking machine	Squirrel cage/ constant	B	Varying, 1 hour	SPT, SPTFC, WTRTT, WTRTTFC	C	Starting	MAG, LVP	Nonautomatic	WTRTT	Local PB	
Sliding padeye	Squirrel cage/ constant	D	Intermittent	WTRTT	A	Starting, reversing	MAG, LVP	Nonautomatic	DP	Remote PB (WTRTT)	
Stairway, electric	Squirrel cage/ constant	B	Continuous	DPP	A	Starting, reversing	MAG, LVP	Nonautomatic	DP	Remote PB	Limit switch and interlocks as required

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Steering gear:											
Fill and drain pumps	Squirrel cage/ constant	B	Contin-uous	DPP, TEFC	A	Starting	MAG, LVP	Nonauto-matic	DP	Local PB	
Main pumps	Squirrel cage/ constant	C	Varying, 1/2 hour	DPP	A	Starting	MAG, LVRLSE	Nonauto-matic	DP	Remote mater, emergency run PB	Selector switch as required
Servo pump	Squirrel cage/ constant	B	Contin-uous	DPP	A	Starting	MAG, LVP or LVRLSE	Nonauto-matic	DP	Remote PB	Selector switch as required
Towing machine	As re-adjust-able, varying	D	Varying, 1 hour	WTRTT	A	Starting, reversing speed selection	MAG, LVP	Nonauto-matic	DP	Remote drum	
Turbine gland exhauster	Squirrel cage/ constant	B	Contin-uous	TE, TEFC	A	Starting,	Manual	-	DP	-	
Turning gear	Squirrel cage constant	D	Contin-uous	DPP	A	Starting, reversing	MAG, LVP	Nonauto-matic	DP	Local PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Turntable, vehicle	Squirrel cage/ varying	C	Contin-uous, short time 15 minute	TE, TEFC	A	Starting, reversing	MAG, LVP MAG, LVP	Nonauto-matic	WTRTT	Remote PB	Limit switch and interlocks as required
Valves, motor operated	Squirrel cage/ constant	B	Contin-uous	TE, WTRTT	A	Starting, reversing	MAG, LVP	Automatic and non-automatic	DP, TE	Local, remote PB	MIL-V-17501
Winches:											
Bathythermo-graph	Squirrel cage/ multispeed	B	Short time, 1/2 hour	WTRTT	A	Starting, speed, selection	MAG, LVP	Nonauto-matic	WTRTT	Remote drum	MIL-W-19882
Constant tension	Squirrel cage/ constant	D	Contin-uous	WTRTT	A	Starting, reversing	MAG, LVP	Nonauto-matic	DP	Remote PB (WT)	
Drum (boat davit)	Squirrel cage/ constant	D	Short time, 1/4 hour	SPT, WTRTT	A	Starting, reversing	MAG, LVRLSE	Nonauto-matic	WTRTT	Remote drum	Drum switch is spring return to "OFF"
Drum (electric)	Squirrel cage/ multispeed	D	Varying, 1/2 hour	SPT, WT	A	Starting, reversing speed selection	MAG, LVRLSE	Nonauto-matic	WTRTT	Remote drum	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclosure	Navy service	Function	Operation protection	Performance	Enclosure		
Winches (continued):											
Drum (hydraulic)	Squirrel cage/ constant	B	Continuous	WTRTT	A	Starting	MAG, LVP	Nonautomatic	DP	Local PB	MIL-W-17265
Drum (magnetic clutch)	Squirrel cage/ varying	B	Varying, 1/2 hour	SPT, WTRTT	A	Starting	MAG, LVP	Nonautomatic	WTRTT	Remote PB	
Gypsy	Squirrel cage/ multispeed	D	Full load 1/2 hour, other, continuous	WTRTT	A	Starting reversing speed selection	MAG, LVRLSE	Nonautomatic	DP	Remote drum (WTRTT)	MIL-W-15802
Main deck ramp	Squirrel cage/ constant	B	Short time	DPP	A	Starting, reversing	MAG, LVP	Nonautomatic	DP	Remote PB	Limit switch and interlocks as required
Vehicle amphibian receiver	Squirrel cage/ constant	B	Continuous	DPP	A	Starting	MAG, LVP	Nonautomatic	DP	Remote PB	

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TABLE I. Ac motor and controller application requirements. - Continued

Auxiliary name	Motor requirements					Controller requirements				Master switches	Remarks and applicable specifications for auxiliaries 1/
	Type/ speed	Ac motor design	Duty	Enclo-sure	Navy ser-vice	Function	Operation protection	Perfor-mance	Enclo-sure		
Windlass anchor:											
Electric	Squirrel cage/ constant	B	Varying	DPP	A	Starting	MAG, LVP	Nonauto-matic	DP	Remote PB (WTRTT)	
Electro-hydraulic	Squirrel cage/ constant	B	Varying, 1 hour	DPP	A	Starting	MAG, LVP	Nonauto-matic	DP	Remote PB	
Replenishment pump	Squirrel cage/ constant	B	Contin-uous	DPP	A	Starting	MAG, LVP	Nonauto-matic	DP	Remote PB	
Window wipers	Permanent magnet/ adjustable		Contin-uous	Open or WTRTT	C	Starting, speed selection	Manual OL	-	DP	-	MIL-W-3459

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-2189(SH) SECTION 302-1	DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS AC MOTOR AND CONTROLLER APPLICATION REQUIREMENTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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